Harshit Singh

Data Engineer | ETL Pipelines • Data Infrastructure • Distributed Systems • Cloud & Databases

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Education

Vellore Institute of Technology, Chennai — Chennai, India
*B.Tech, Electronics and Computer Engineering — CGPA: **7.97*** — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
*Class XII: **87.16%** — Class X: **89.83%*** — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] [**GitHub**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Built a time-series data validation pipeline processing **15 Hz** Extended Kalman Filter telemetry against raw encoder logs, quantifying positional/orientation drift and measuring a **60% reduction** in mean spatial error (**0.45 m → 0.18 m**) and **69.2%** in angular error across 10 m ground-truth trials.
- Engineered a windowed spatial-filtering pipeline to ingest and process high-frequency LiDAR sensor streams (∼**3,600 points/sec** at **10 Hz**), achieving **95% obstacle-detection accuracy** across 20 randomized stress-test trials with **sub-50 ms** end-to-end telemetry latency.

Projects

SPARC — Streaming Data Pipeline *(Python, Pandas, NumPy)* [**Demo**] [**Web-App**]

- Engineered a streaming ETL pipeline transforming **921,600-dim** video frames into a **229-dim** feature schema with **sub-8ms** latency at **92.4% accuracy**.
- Cut false-positive ingestion by **92%** across 41 classes via sliding-window filtering — **Top 5** national finalist (500+ teams), SIH 2025. [**Certificate**]

TRIDENT — Telemetry Pipeline & Data Infra *(Flask, SQLite, Docker)* [**Live**] [**Demo**] [**GitHub**]

- Automated ingestion of **17,500+** hourly telemetry records via a Python ETL pipeline querying the Meteostat API, with outlier filtering and feature scaling.
- Designed a relational SQLite schema with indexed keys and FK constraints, containerizing the backend via Docker/Gunicorn — cutting image size **80%**, sustaining **sub-2s** latency across 15+ REST routes.

Weather Risk AI — ETL & Forecasting Pipeline *(Pandas, Scikit-Learn, LSTM)* [**GitHub**]

- Built an automated ETL pipeline ingesting **1,460+** daily records across 6 features, applying ffill/bfill imputation and leakage-safe MinMaxScaler transforms.
- Transformed data into **(N,10,6)** sliding-window tensors feeding a 2-layer LSTM (**sub-0.05 MSE**), modeling flood risk across 8 zones via graph diffusion.

GNOME Stocks Suite — Data Pipeline & Caching Infra *(Python, Flask, SQLite)* [**Live**] [**GitHub**]

- Engineered a batch ingestion pipeline filtering **152,000+** records to ∼**2,500** tradeable assets via **/dev/shm** indexing, cutting query latency **99.3%** (800ms → <5ms).
- Architected a symbol normalization router unifying 3 APIs into one schema, backed by a tiered SQLite + TTL cache and cron keep-alive, sustaining **99.9%** uptime with 0s cold-starts.

Skills

Languages: Python, SQL, C, Java
Data Engineering: ETL Pipelines, Batch & Stream Processing, Data Ingestion, Caching, REST APIs, Schema Design
Data Infrastructure: PostgreSQL, MySQL, SQLite, Docker, Gunicorn, Cloud Deployment (Render, AWS), GitHub Actions (CI/CD)
Data Tools: Pandas, NumPy, Matplotlib, NetworkX, Flask, Git
ML & Time-Series: Scikit-Learn, TensorFlow/Keras, LSTM, Time-Series Forecasting
Professional Skills: Data-Driven Problem Solving, Cross-Functional Collaboration, Communication, Time Management

Achievements & Leadership

- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; **INR 15,000** award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; **INR 5,000** award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting efficiency by **35%** and securing **2 sponsors.** [**LOA**] [**Certificate**] [**LOR**]

Languages

English (Fluent) | **Hindi** (Native)